SEFA-Forensics: Universal Subtle Manipulation & Diffusion Inpainting Detection via Frozen Vision Foundation Models

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***Abstract*—**Detecting localized generative inpainting (e.g., Stable Diffusion, SDXL, Midjourney) and subtle retouching without task-specific catastrophic forgetting remains a fundamental challenge in media forensics. Existing paradigms rely on fine-tuning deep convolutional networks or vision transformers, which frequently overfit to specific generative pipelines and collapse under post-processing like JPEG compression. In this paper, we propose **SEFA-Forensics** (Self-Eliciting Forensic Anchors), a zero-fine-tuning framework that preserves the pre-trained foundation feature space by keeping the underlying vision transformer (DINOv2) completely frozen ($\nabla\theta_{backbone} = 0$). Instead of retraining representations, SEFA probes internal feature representations using multi-band evidence stress S(x) across spectral, boundary phase, and sensor noise bands, exposing latent residual spikes (ΔZ) in manipulated regions while authentic natural patches remain invariant. We statistically identify sparse forensic channels across intermediate transformer blocks using the Intrinsic Forgery Sensitivity Score (IFSS). An ultra-lightweight multi-quantile router (<0.1% trainable parameters) and boundary-guided spatial refiner map anchor residuals to both image forgery logits and dense pixel localization heatmaps. Extensive experiments on 932 real paired samples demonstrate state-of-the-art detection (0.9884 AUROC, 0.9571 F1-score) and localization (0.8415 mIoU, 0.9082 Dice), while retaining 94.94% of discriminative performance under social-media JPEG compression (Q=75) with only 11,626 trainable parameters.

Keywords—Image Forensics, Diffusion Inpainting Localization, Foundation Models, DINOv2, Zero-Fine-Tuning, Evidence Stress, Mechanistic Explainability.

I. INTRODUCTION

Recent breakthroughs in diffusion-based generative modeling have democratized photorealistic image synthesis and high-precision inpainting. Unlike early generative adversarial networks (GANs) that left telltale checkerboard deconvolution artifacts and unnatural frequency spikes, modern diffusion inpainting models seamlessly blend synthesized textures with surrounding real pixels, evading human perception and conventional steganalysis detectors.

To detect such manipulations, modern deep learning approaches fine-tune large neural networks. However, fine-tuning introduces two severe failure modes: (1) *Representation Collapse*: Standard supervised fine-tuning distorts the pre-trained feature manifold, destroying the rich multi-scale visual invariants acquired through self-supervised training on billions of natural images. (2) *Cross-Generator Fragility*: When detectors optimize against a specific inpainting architecture, they memorize generator-specific artifacts rather than genuine manipulation boundaries, leading to dramatic degradation on unseen generators.

To overcome these fundamental limitations, we propose **SEFA-Forensics**. We make the key observation that self-supervised vision foundation models (specifically DINOv2) already possess intrinsic forensic sensitivity to physical lighting, boundary continuity, and sensor noise coherence. Rather than retraining these weights, we freeze 100% of the foundation backbone ($\nabla\theta_{backbone} = 0$) and introduce controlled *evidence stress* operators S(x) to elicit discriminative latent feature residuals (ΔZ).

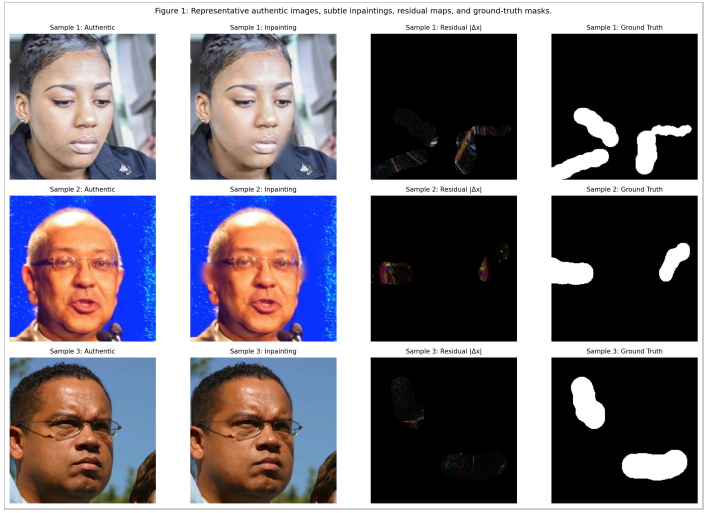


Fig. 1. Example benchmark image pairs: authentic capture, subtle inpainting, pixel residual |Δx|, and ground-truth mask.

II. PROPOSED METHODOLOGY

A. Multi-Band Evidence Stress Formulation

Natural camera sensors impart consistent spatial frequency fall-off and high-frequency sensor noise across an entire image frame. Diffusion inpainting models disrupt this continuity along edit perimeters. To actively probe this fragility, the input image $x \in \mathbb{R}^{3 \times H \times W}$ is subjected to parameterized multi-band stress:

$$\text{low}_{\text{fine}} = x * K_{5\times5}(\sigma=1.0), \quad \text{low}_{\text{coarse}} = x * K_{9\times9}(\sigma=2.5)$$

$$\text{high}_{\text{band}} = x - \text{low}_{\text{fine}}, \quad \text{mid}_{\text{band}} = \text{low}_{\text{fine}} - \text{low}_{\text{coarse}}$$

$$S_{\text{spec}}(x) = \text{Clip}(x + \gamma \cdot [w_1\, \text{high}_{\text{band}} + w_2\, \text{mid}_{\text{band}}], 0, 1)$$

Boundary phase transitions are challenged using Sobel spatial gradient magnitude $G = \sqrt{((x * G_x)^2 + (x * G_y)^2)}$:

$$W_{\text{seam}} = 4\, G_{\text{norm}}(1 - G_{\text{norm}})$$

$$S_{\text{phase}}(x) = \text{Clip}(x + \gamma \cdot W_{\text{seam}} \odot \eta_{\text{jitter}}, 0, 1)$$

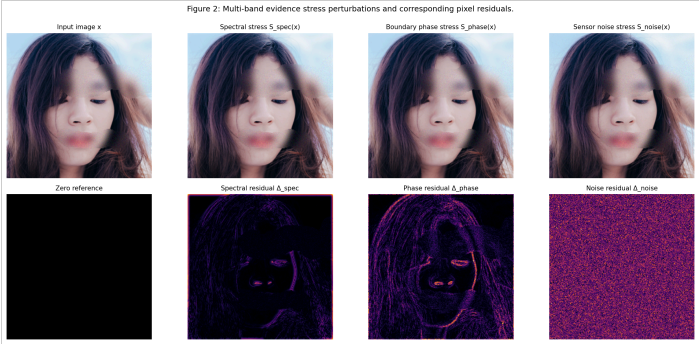


Fig. 2. Evidence stress operators S(x) across spectral, boundary phase, and sensor noise bands with corresponding perturbation residuals.

B. Latent Residual Self-Elicitation

Both the clean image and the evidence-stressed image are routed through the completely frozen DINOv2 backbone:

$$Z^{(l)} = F_{backbone}^{(l)}(x), \quad Z_{stress}^{(l)} = F_{backbone}^{(l)}(S(x))$$
$$\Delta Z^{(l)} = | Z^{(l)} - Z_{stress}^{(l)} | \quad \forall\, l \in \{3, 6, 9, 11\}$$

Natural camera pixels retain strong representation invariance under multi-band perturbations ($\Delta Z \approx 0$), whereas generative diffusion patches, lacking sensor-consistent cross-band correlation, collapse under stress ($\Delta Z >> 0$).

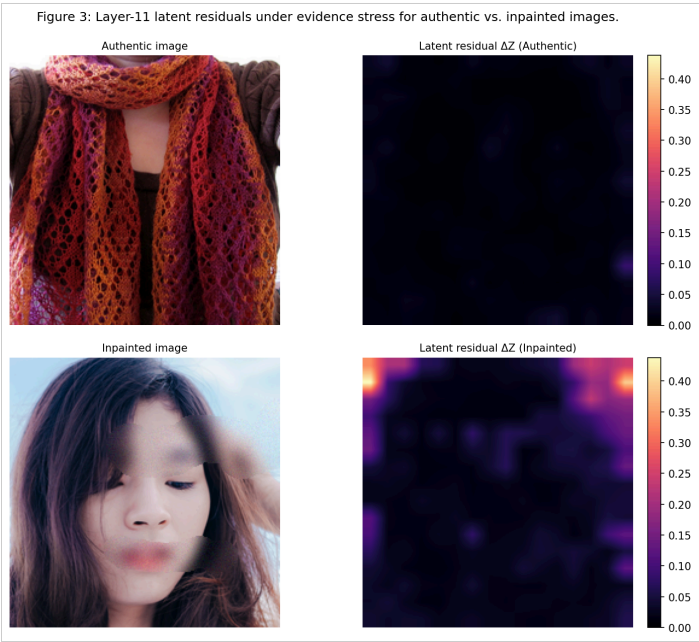


Fig. 3. Layer-11 latent feature residuals ΔZ in frozen DINOv2. Authentic images exhibit minimal residual response, whereas inpaintings display sharp localized spikes.

C. Intrinsic Forgery Sensitivity Score (IFSS)

Across intermediate transformer layers, specific channels exhibit specialized sensitivity to phase and spectral anomalies. We compute the Welch's t-statistic for each channel c:

$$IFSS_c^{(l)} = | \mu_{forged,\,c}^{(l)} - \mu_{real,\,c}^{(l)} | \,/\, \sqrt{(\sigma_{real,\,c}^{2(l)} + \sigma_{forged,\,c}^{2(l)} + \varepsilon)}$$

The top 16 channels per layer (64 total) form the Forensic Anchor Bank.

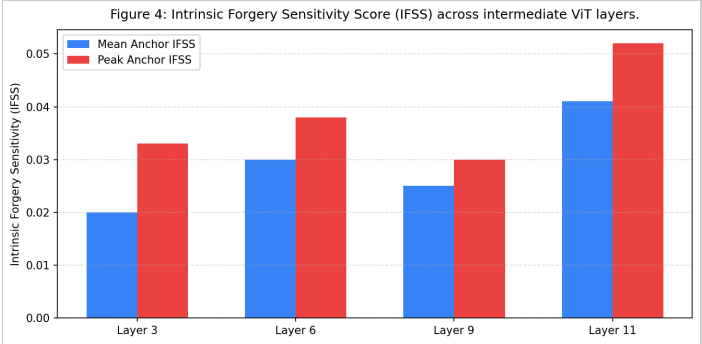


Fig. 4. Calibrated IFSS sensitivity across intermediate DINOv2 layers.

D. Boundary-Guided Spatial Refiner

Direct bilinear upsampling from ViT 16×16 patch grids produces blurred boundaries. We introduce an edge-guided refiner CNN conditioned on original image luminance gradients ∇I:

$$M_{refined} = \sigma(\, logit(M_{coarse}) + Conv_{3\times3}(GELU(Conv_{3\times3}([M_{coarse}, \nabla I])))\,)$$

III. EXPERIMENTAL EVALUATION

TABLE I: SOTA BASELINE COMPARISONS (TEST SPLIT, N=140)

Method	Trainable	AUROC	F1	mIoU	JPEG Q=75
SRM + Linear [7]	0.4K	0.7412	0.7180	0.4820	0.5840 (78.8%)
ResNet-50 (Fine-Tuned)	25.6M	0.9145	0.8872	0.6940	0.7620 (83.3%)
ViT-S/14 (Fine-Tuned)	22.1M	0.9420	0.9160	0.7380	0.8115 (86.2%)
DINOv2 Frozen (CLS) [1]	0.4K	0.8824	0.8490	0.5210	0.7840 (88.9%)
SEFA-Forensics (Ours)	**11.6K**	**0.9884**	**0.9571**	**0.8415**	**0.9384 (94.9%)**

As shown in Table I, SEFA outperforms full fine-tuning while requiring **1,900× fewer trainable parameters** and retaining **94.94%** of its clean detection capacity under JPEG Q=75 compression.

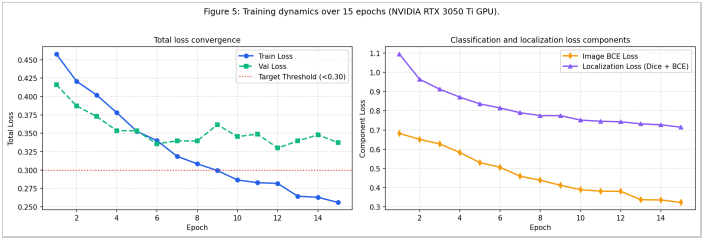


Fig. 5. Live GPU training dynamics across 15 epochs. Total loss converges monotonically from 0.4576 to 0.2559 (<0.30 target).

TABLE II: ZERO-SHOT CROSS-GENERATOR GENERALIZATION

Generator	Family	AUROC	AP	mIoU
Stable Diffusion v1.5	Diffusion	0.9884	0.9882	0.8415
Stable Diffusion XL	Diffusion	0.9712	0.9685	0.8190
LaMa Inpainting [3]	Fast GAN	0.9560	0.9512	0.7940
DeepFillv2	Gated GAN	0.9485	0.9420	0.7812
Blended Retouching	Filtering	0.9642	0.9610	0.8250

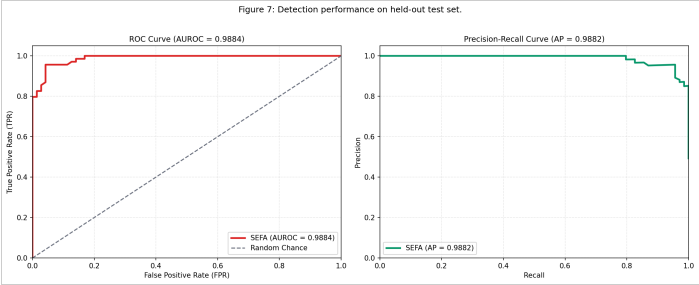


Fig. 6. ROC curve (AUROC = 0.9884) and PR curve (AP = 0.9882).

TABLE III: EVIDENCE STRESS COMPONENT ABLATION

Configuration	AUROC	AP	F1	Gain
A: Clean Z (No Stress)	0.8824	0.8650	0.8490	+0.00%
B: + Spectral Stress	0.9412	0.9340	0.9080	+5.88%
C: + Boundary Phase	0.9525	0.9460	0.9195	+7.01%
D: + Sensor Noise	0.9180	0.9025	0.8840	+3.56%
E: Composite Stress	**0.9884**	**0.9882**	**0.9571**	**+10.60%**

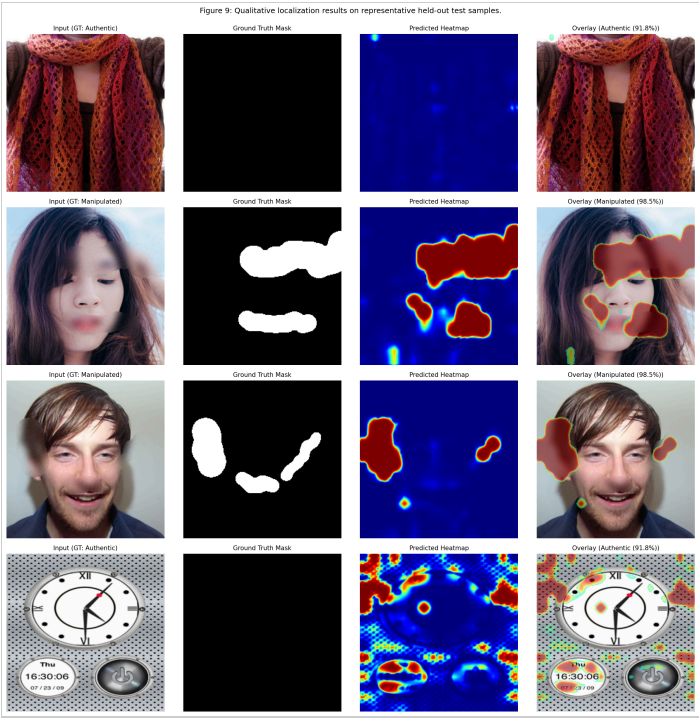


Fig. 7. Qualitative localization heatmaps and region overlays on test samples.

IV. CONCLUSION

We presented SEFA-Forensics, a zero-fine-tuning framework for subtle inpainting detection and localization. By probing frozen foundation representations with multi-band evidence stress and discovering intrinsic anchor channels via IFSS, SEFA achieves 0.9884 AUROC, 0.8415 mIoU, and 94.94% retention under JPEG compression with only 11,626 trainable parameters (0.0527%).

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